Multiscale Symmetric Difference Attention-Based Transformer for Panchromatic and Multispectral Classification

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***Abstract*—Nowadays, effectively exploiting multispectral (MS) and panchromatic (PAN) imagery for their joint classification task has attracted wide attention. However, due to their heterogeneous resolutions and distinct modality characteristics, the precise alignment of different-scale semantic features and the compact fusion of modality-specific information remain challenging. To tackle these issues, we propose a multiscale symmetric difference attention-based Transformer (MSAFormer). It comprises two key modules: the texture-aware multiscale feature generation (TMFG) module and the symmetric difference feature divergence (SDFD) interaction module. By exploiting PAN's high-resolution texture to bridge the inherent texture deficiency of MS at continuous resolutions, TMFG builds a multiscale space with precise texture alignment. This module constructs a robust foundation for the comprehensive extraction of dual-source-specific features during interaction. Subsequently, by applying the concept of symmetric difference between sets to the aligned dual-modality groups of different scales, SDFD generates compact and comprehensive multiscale fusion representations in a nondecoupled manner. These lightweight fusion features integrate the enhanced modality-specific features and eliminate the overlapped redundancy of shared information, ultimately yielding superior classification performance. Several experimental results validate the effectiveness and robustness of the network. Codes are available at https://github.com/Xidian-AIGroup190726/MSAformer**

***Index Terms*—Inverted attention, multiscale, multisource classification, panchromatic (PAN) and multispectral (MS), Transformer.**

I. Introduction

With the rapid development of geospatial information technology, remote sensing satellites have made it easier to acquire images from diverse sources, such as multispectral (MS) and panchromatic (PAN) images [1], [2], [3], [4]. The use of single-modal data can no longer meet the expected performance requirements. Consequently, researchers have increasingly focused on improving the classification accuracy of remote sensing images through multisource data fusion, particularly the fusion of PAN and MS data.

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These two data sources provide complementary information: PAN data offer high spatial resolution but low spectral resolution, whereas MS data provide high spectral resolution but low spatial resolution. Effectively combining the strengths of PAN and MS data can significantly enhance image classification performance.

Most PAN and MS image fusion classification algorithms can be categorized into two types: traditional fusion methods focusing on data-level fusion and deep learning fusion methods focusing on feature-level fusion [5], [6].

Traditional data-level fusion approaches typically apply pansharpening to MS images before classification, aiming to integrate the rich spectral content of MS with the high spatial details of PAN [7], [8]. Existing pansharpening techniques are mainly categorized into three groups: component substitution (CS), multiresolution analysis (MRA), and variational optimization (VO). CS-based methods (e.g., PCA [9], GS [10], and IHS [11]) replace MS spatial components with PAN information in a transformed domain. MRA approaches (e.g., wavelet transform and Laplacian pyramid [12]) extract and fuse multiscale features from both modalities. VO-based methods formulate an energy function and solve it through observation models or sparse representations [13]. However, these approaches may introduce spectral distortion.

With rapid progress in deep learning [14], [15], [16], deep-learning-based methods feed multisource data into neural networks to extract high-level features. Li et al. [17] use deep CNNs to extract pixel-block pair features and apply decision fusion for final classification. Gao et al. [18] propose an unsupervised network with an attention-embedded autoencoder for multispectral-aided hyperspectral image super-resolution. Other studies on domain adaptation, large-scale anomaly detection, and image captioning have also widely employed deep learning and specifically address the unique characteristics of remote sensing data [19], [20], [21].

Nowadays, following the widespread adoption of Transformers in natural language processing, they are gradually applied to the field of vision [22]. Ma et al. [23] utilize cross-domain long-range learning and Swin Transformer, preserving image structure, details, and appropriate intensity. Hong et al. [24] propose a Transformer-based backbone that models spectral sequences with group-wise embeddings and preserves information via cross-layer skip connections.

Multiscale feature extraction is crucial in remote sensing classification [25], as land covers vary in size and spatial patterns. Traditional approaches, such as resampling and image

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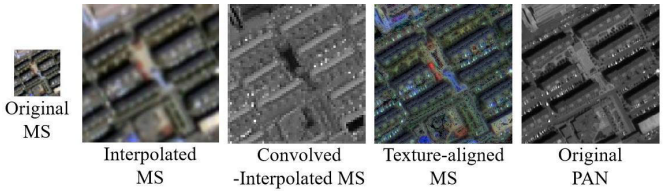

Fig. 1. Illustration of motivation 1). In the middle, the MS images use three different upsampling methods. Conventional interpolation and convolution approaches result in blurry representations of MS. Many details are misaligned, and small color patches become indistinct. Conversely, the texture-aligned MS preserves details and matches the sharpness of PAN.

pyramids, capture multiresolution features [26], [27], but deep learning has enabled more adaptive strategies. Zhong et al. [28] propose a multiscale CNN for hierarchical feature learning, while Shi et al. [29] design a dual-branch Transformer to fuse local spectral and global contextual features. Recent works increasingly focus on adaptive scale selection and efficient fusion within Transformer frameworks.

In remote sensing bimodal image fusion classification, several advanced methods have shown promising results and are used as comparison baselines. NNCNet [30] utilizes nearest neighbor-based contrastive learning to enhance multisource feature alignment; MFT [31] leverages multimodal information in the Transformer encoder to achieve better generalization; AM3Net [32] performs a mutual-guided module to achieve feature collaboration; CRHFF [33] fuses cross-resolution hidden layer features from MS-PAN images to integrate global–local information; and GCFNet [34] adopts multiscale cross-modal branches and adaptive decision fusion to improve MS-PAN classification. In more recent studies, ISSP-Net [35] introduces a pixel-guided spatial module and a time–frequency spectral module to extract discriminative frequency-domain information. SNCF-Net [36] proposes a network based on continuous-scale inputs and response convolution kernel to achieve feature interaction. Although these methods have achieved promising results, some of them still suffer from early stage resolution misalignment and the risk of losing fine-grained texture information, while others rely on a separation-fusion strategy that may introduce redundancy.

These limitations motivate us to further explore more effective multiscale alignment and compact fusion strategies.

1) To better extract multiscale features from modalities with heterogeneous resolutions, many studies adopt conventional resampling techniques such as stride convolution and pooling. These operations lack fine-grained design, which may lead to the loss and discontinuity of high-resolution texture and cause misalignment of PAN and MS as shown in Fig. 1. Such early stage errors can be unpredictably amplified during the network's evolution. Namely, these errors further hinder the downstream exploitation of dual-source-specific features, which may lead to misclassification for land covers with only subtle spectral differences.
2) To leverage the complementary nature of MS and PAN, many studies explicitly decouple specific and shared features and then concat them for classification. However, this clean separation often comes at a cost. The aggregate information dimensionality retained after decoupling is frequently greater than that before decoupling, which may lead to information dispersion and latent redundancy, thereby diminishing the benefits of the decoupling process.

Based on the analysis of the aforementioned issues, we propose a multiscale symmetric difference attention-based Transformer for PAN and MS classification. Our main contributions are given as follows.

1) We propose a texture-aware multiscale feature generation (TMFG) module, which progressively builds a multiscale space with precise texture alignment. By exploiting PAN's high-resolution texture to bridge the inherent texture deficiency of MS at continuous resolutions, the two modalities achieve not only resolution alignment but also precise, detailed semantic alignment in multiscale space. This module constructs a robust foundation for the comprehensive extraction of dual-source-specific features during feature interaction.
2) We propose a symmetric difference feature divergence (SDFD) interaction module, which generates compact and comprehensive multiscale fusion representations across modalities. Leveraging the signal response principle, the module applies the concept of symmetric difference between sets to the aligned dual-modality groups of different scales in a nondecoupled manner. This mechanism obtains a series of relatively lightweight multiscale fusion features that highlight modality-specific features and eliminate the overlapped redundancy of shared information.

Algorithm 1 Multihead Self-Attention Mechanism in ViT

Input: Feature tensor $x \in \mathbb{R}^{B\times N\times C}$
Output: Output tensor $x' \in \mathbb{R}^{B\times N\times C}$
1) Compute $qkv = \text{Linear}(x)$
2) Reshape qkv to shape $(B, N, 3, h, d)$, where $h = \text{num_heads}$ and $d = C/h$
3) Split qkv into q, k, v
4) Normalize q and k (optional)
5) Scale queries: $q \leftarrow q/\sqrt{d}$
6) Compute attention weights: $A = \text{softmax}(q \cdot k^T)$
7) Compute attention output: $z = A \cdot v$
8) Concatenate heads: reshape z to shape (B, N, C)
9) Project output: $x' = \text{Linear}(z)$

II. Related Work

A. Vision Transformer

The vision Transformer (ViT) was first proposed by Dosovitskiy et al. [22], attempting to apply the Transformer to computer vision tasks by treating images as language sequences. ViT first divides the input image into small patches and projects them into fixed-length vectors. These vectors then go through position embeddings and are put into multilayer self-attention mechanisms to extract contextual dependencies. The multihead self-attention mechanism is performed as Algorithm 1.

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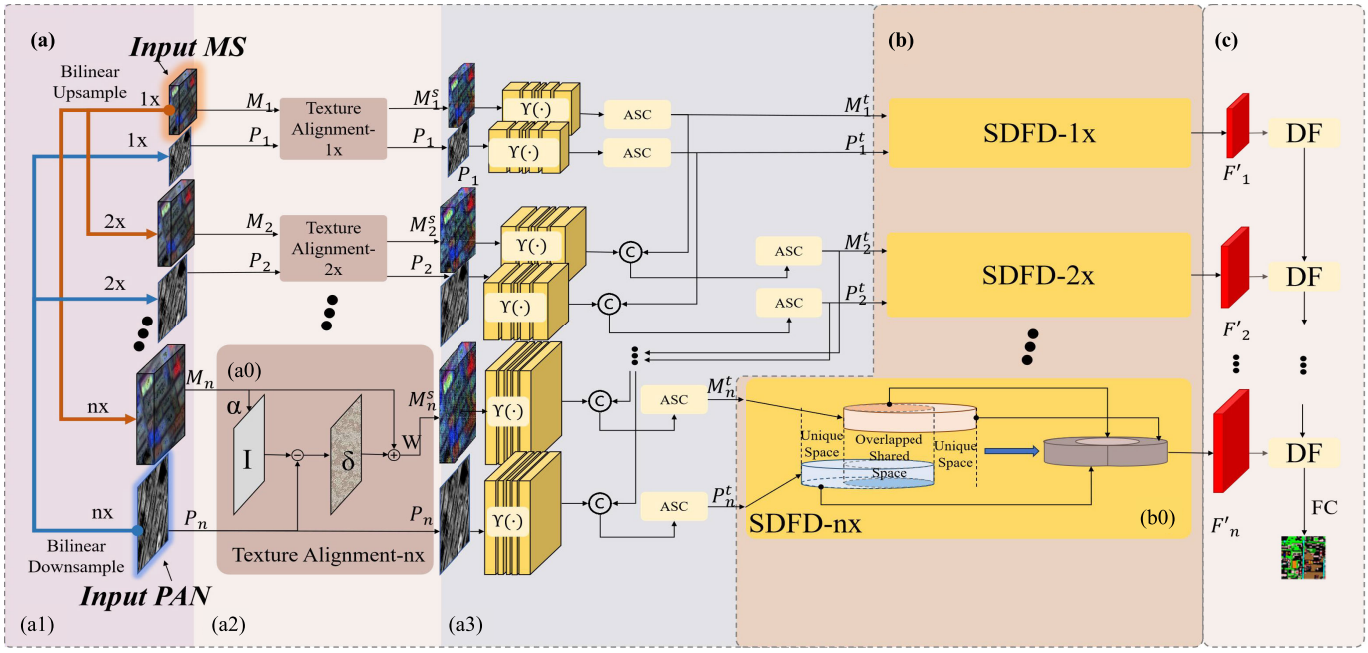

Fig. 2. Illustration for the multiscale symmetric difference attention-based Transformer. The network is composed of three core sections. (a) Data processing architecture centered around the TMFG module. (a0) Detailed depiction of texture alignment. (a1)–(a3) Components of (a). (b) Feature interaction architecture centered around the SDFD module. (b0) Schematic of SDFD-nx. (c) Fusion classification based on decoupling. "ix" in (a) and (b) stands for the resolution size of the ith group.

In this implementation, multihead attention is realized by reshaping and splitting along the head dimension. Each head operates on a separate slice of the projected feature space, allowing attention computations to be performed in parallel. However, since all heads originate from a single linear mapping, this design constitutes a form of structured block-wise parallelism rather than truly independent feature learning, potentially limiting the diversity of captured attention patterns. Our proposed method retains the conventional design while introducing targeted modifications to increase the diversity of each attention head.

B. Body-Edge Decoupled Fusion

Li et al. [37] proposed a body-edge decoupled supervision strategy for semantic segmentation. By constructing flow-field-based body representations and deriving edge features through residual learning, their method effectively enhances both semantic consistency and boundary accuracy. Several subsequent studies have also adopted similar decoupling strategies in different segmentation tasks [38], [39]. Motivated by this idea, our proposed method adopts a similar decoupling concept in the feature fusion stage to reduce body-edge interference.

III. Proposed Method

In this section, we introduce the entire process of the multiscale symmetric difference attention-based Transformer (MSAFormer), as shown in Fig. 2. We denote MS as $M \in \mathbb{R}^{H\times W\times 4}$ and PAN as $P \in \mathbb{R}^{4H\times 4W\times 1}$. The network architecture comprises three key parts: data processing centered around the TMFG module, feature interaction centered around the SDFD module, and fusion classification based on decoupling. These components are elaborated in Sections III-A–III-C.

A. Texture-Aware Multiscale Feature Generation

To achieve fine-grained alignment of different-scale semantic features, we propose a TMFG module. TMFG is composed of three parts, including the construction of a hierarchical multiresolution representation, texture alignment, and cross-scale feature communication. The overall architecture of TMFG is shown in Fig. 2(a). Components correspond to (a1)–(a3).

1) Construction of Hierarchical Multiresolution Representation: To model the diverse interactions of features across different scales, we construct hierarchical representations of multiresolution feature groups with preliminary resolution alignment

$$\begin{aligned} M_1 &= M, P_1 = \text{DownBilinear}_1(P) \\ M_i &= \text{UpBilinear}_i(M), P_i = \text{DownBilinear}_i(P) \\ M_n &= \text{UpBilinear}_n(M), P_n = P \end{aligned} \tag{1}$$

where different $i \in \{1, 2, \ldots, n\}$ denotes the upsample or downsample bilinear interpolation for different groups. M_i and P_i share the exact resolution. As i increases, their resolution increases accordingly. This nonlearning method preserves the authenticity of the data and thus helps construct the texture-aware network. Then, we get a series of dual-modality input feature groups of multiple resolutions

$$\begin{aligned} group_1 &= \left\{M_1 \in \mathbb{R}^{H\times W\times 4},\ P_1 \in \mathbb{R}^{H\times W\times 1}\right\} \\ group_i &= \left\{M_i \in \mathbb{R}^{i(4H)\times i(4W)\times 4},\ P_i \in \mathbb{R}^{i(4H)\times i(4W)\times 1}\right\} \\ group_n &= \left\{M_n \in \mathbb{R}^{4H\times 4W\times 4},\ P_n \in \mathbb{R}^{4H\times 4W\times 1}\right\} \end{aligned}$$

where $_i(\cdot)$ represents a scale distribution function that maps a variable to the range $[1/4, 1]$ of its original magnitude. A larger i corresponds to a weaker scaling effect. The resolution scale ranges from $H \times W$ to $4H \times 4W$, falling between the input MS

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and PAN, which preserves the integrity and consistency of both modalities. Finally, it constructs a hierarchical multiresolution foundation across consistent scales, providing a resolution-adaptive series that supports both coarse and fine geometric recognition.

2) Texture Alignment: To extract features from modalities with heterogeneous resolutions, many studies adopt simple techniques, such as stride convolution and pooling, to resample MS to match PAN. Such sampling merely enlarges the low-resolution MS image without introducing new structural information, resulting in a larger but blurrier representation. Essentially, it fails to align their inherent differences in semantic details and may cause detail loss and discontinuity.

To achieve semantically aligned multiscale representations, we adopt a multiscale IHS alignment strategy. While IHS and its variants have been extensively employed in pansharpening applications [40], their extension toward feature alignment across scales represents a novel attempt. Leveraging the fine-grained spatial details of the PAN image, the textures of MS images are enriched and aligned to PAN.

Based on the hypothesis that spectral distortion can be reduced by approximating the intensity component I as closely as possible to PAN, for M_i in $group_i = \{M_i, P_i\}$

$$I_i = \sum_{c=1}^{4} \alpha_c M_{i_c} \tag{2}$$

where c represents the channel index of M_i. α_c is an adaptive function obtained through the following constraint equation:

$$\arg_\alpha \min F(\alpha) = \sum_{c=1}^{4} \left(\alpha_c M_{i_c} - P_i\right)^2 + \gamma \sum_{c=1}^{4} \left(max\left(0, \alpha_c\right)^2\right). \tag{3}$$

Then, edge detection operators are introduced to better extract and integrate edge features

$$W_{i_c} = \frac{M_{i_c}}{\frac{1}{n}\sum_{c=1}^{n} M_{i_c}} \left(\beta W_{M_{i_c}} + (1-\beta) W_{P_i}\right) \tag{4}$$

where β is a hyperparameter that is set to 0.5 after manual tuning. $W_{M_{i_c}}$ and W_{P_i} denote the edge weights for M_{i_c} and P_i, respectively, which are derived from the computation of edge detection factors. Furthermore, each layer of the texture-aligned spectral image M_i^s is obtained as follows:

$$M_{i_c}^s = M_{i_c} + W_{i_c}\left(P_i - I_i\right). \tag{5}$$

A schematic visualization of the result is shown in Fig. 1, and the illustration of the computation process is shown in Fig. 2(a0). As a result, we obtain hierarchical input features $group_i' = \{M_i^s, P_i\}$ with not only consistent resolution alignment but also semantic texture alignment. Therefore, a precisely aligned multiscale foundation is prepared for subsequent processing.

3) Cross-Scale Feature Communication: To bridge the spatial–channel semantic gap across resolutions, we apply cross-scale communication before feeding the constructed multiscale groups into the interaction network.

Intragroup Enhancement: First, each feature group is processed by the spatial–channel attention mechanism (ASC) to refine both spatial locality and spectral discriminability. For $group_i' = \{M_i^s, P_i\}$

$$\begin{cases} M_i^{s'} = \text{concat}\left(\Upsilon\left(M_i^s\right), S_{M_{i-1}}\right) \\ \text{Map-S}\left(M_i^{s'}\right) = \mathcal{C}_{d\to d/r_1\to 1}\left(M_i^{s'}\right) \\ \text{Map-C}\left(M_i^{s'}\right) = \mathcal{C}_{d\to d/r_2\to d}\left(\text{AvgPool2d}\left(M_i^{s'}\right)\right) \\ M_i^{s''} = M_i^{s'} + M_i^{s'} \odot \sigma\left(\text{Map-S}\left(M_i^{s'}\right)\right) \\ M_i^t = M_i^{s''} + M_i^{s''} \odot \sigma\left(\text{Map-C}\left(M_i^{s'}\right)\right). \end{cases} \tag{6}$$

In the same way, P_i is operated as follows:

$$\begin{cases} P_i' = \text{concat}\left(\Upsilon\left(P_i\right), S_{P_{i-1}}\right) \\ \text{Map-S}\left(P_i'\right) = \mathcal{C}_{d\to d/r_1\to 1}\left(P_i'\right) \\ \text{Map-C}\left(P_i'\right) = \mathcal{C}_{d\to d/r_2\to d}\left(\text{AvgPool2d}\left(P_i'\right)\right) \\ P_i'' = P_i' + P_i' \odot \sigma\left(\text{Map-S}\left(P_i'\right)\right) \\ P_i^t = P_i'' + P_i'' \odot \sigma\left(\text{Map-C}\left(P_i'\right)\right) \end{cases} \tag{7}$$

where $\Upsilon(\cdot)$ denotes several convolutional layers. $\mathcal{C}_\cdot(\cdot)$ denotes two 1×1 convolutional layers with BN and GeLU. The subscript reflects the variation in dimensional structure. r_1 and r_2 are integers that can divide d. $\odot$ denotes the dot product. $\sigma(\cdot)$ denotes the ReLU. Therefore, Map-S$(\cdot)$ forms the spatial attention map, and Map-C$(\cdot)$ forms the channel attention map. $S_{M_{i-1}}$ and $S_{P_{i-1}}$ are skip connection blocks elaborated in the following text.

Intergroup Communication: To allow low-level fine details and high-level abstract semantics to reinforce each other, intergroup communication is performed to bridge the semantic gap. At each intragroup enhancement, $S_{M_{i-1}}$ and $S_{P_{i-1}}$, the skip connection blocks, are obtained from the dual-modality group of the last smaller resolution

$$S_{M_0} = \varnothing, S_{P_0} = \varnothing \tag{8}$$

$$S_{M_{i-1}} = \text{UpsampleDowndim}\left(M_{i-1}^t\right) \tag{9}$$

$$S_{P_{i-1}} = \text{UpsampleDowndim}\left(P_{i-1}^t\right) \tag{10}$$

where UpsampleDowndim$(\cdot)$ is a set of convolutional blocks for upsampling and reducing the dimension by half. This operation enables suppressing redundant communication contextual flow.

Generally, the three components of TMFG progressively build a multiscale space with precise texture alignment, which provides a robust foundation for the subsequent modality-specific feature interaction.

B. SDFD Interaction

To generate compact fusions of modality-specific information of dual-modality groups across different scales, we designed the SDFD module shown in Fig. 3. For the series of dual-modality features, we perform interaction between two modalities in each group separately.

1) Principle of Inverted Attention: To extract modality-specific and shared information, existing separation-fusion strategies are extensively employed but may lead to expanded and sparse feature representations. Therefore, the module incorporates the concept of symmetric difference into an inverted attention mechanism to distill the three parts of information in a nondecoupled manner. The schematic is

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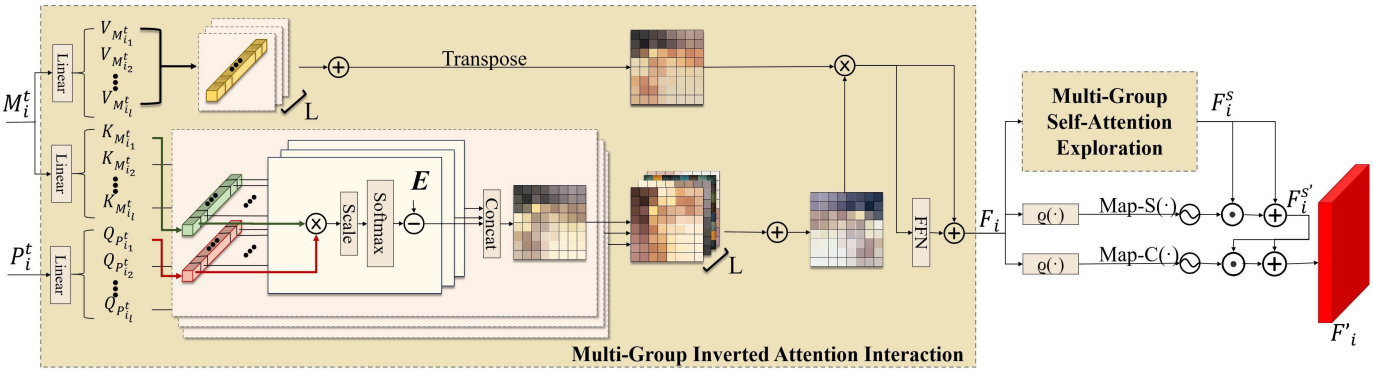

Fig. 3. Illustration of the SDFD-ix module. "ix" stands for the resolution size of the ith group. The multigroup inverted attention interaction and the multigroup self-attention exploration are of the same structure. A detailed elaboration is provided in Section III-B.

shown in Fig. 2(b0). The inverted attention map is calculated as follows:

$$A_{gqk} = E - \text{softmax}\left(\frac{Q_p K_m^T}{\sqrt{d_k}}\right) \tag{11}$$

where E denotes the matrix with all elements equal to one. Since the response between signals with similar distributions is stronger than that between signals with different distributions, the larger weights of traditional attention $Q_p \cdot K_m^T$ represent parts where the signal distributions are similar. Therefore, it can be deduced that when we invert the attention map within its range [0, 1] and get A_{gqk}, the regions with larger weights represent those parts where the signal distributions are different. Thus, as if performing a symmetric difference operation between two modalities, an attention map is generated to simultaneously emphasize specific cues of each modality and compress the overlapped redundant shared features from two parts into one. This core principle is used in the multigroup inverted attention interaction mentioned as follows.

2) Multigroup Attention Strategy: In ViT and its variants, multihead attention operates by dividing the dimensions of the query, key, and value into multiple heads. Each of the heads performs parallel attention computations independently. However, since they all come from the same projection, the pattern captured tends to be uniform. To suppress competition between attention heads and increase the model's diversity, we design a multigroup attention strategy.

Inverted Attention Interaction: For $group_i'' = \{M_i^t, P_i^t\}$

$$m = \left[\text{CLS}; M_{i_1}^t W; M_{i_2}^t W; \ldots; M_{i_N}^t W;\right] + E_{pos} \tag{12}$$

$$p = \left[\text{CLS}; P_{i_1}^t W; P_{i_2}^t W; \ldots; P_{i_N}^t W;\right] + E_{pos} \tag{13}$$

where a learnable CLS token is used for global representation. Position embeddings E_{pos} are added to preserve the sequential order of the input. N denotes the number of patches. W denotes the patch embedding projection. After this

$$Q_p = pG^q = p\begin{bmatrix} G_{11}^q & G_{12}^q & \cdots & G_{1S}^q \\ G_{21}^q & G_{22}^q & \cdots & G_{2S}^q \\ \vdots & \vdots & \vdots & \vdots \\ G_{L1}^q & G_{L2}^q & \cdots & G_{LS}^q \end{bmatrix} \tag{14}$$

$$K_m = mG^k = m\begin{bmatrix} G_{11}^k & G_{12}^k & \cdots & G_{1S}^k \\ G_{21}^k & G_{22}^k & \cdots & G_{2S}^k \\ \vdots & \vdots & \vdots & \vdots \\ G_{L1}^k & G_{L2}^k & \cdots & G_{LS}^k \end{bmatrix} \tag{15}$$

$$V_m = mG^v = m\begin{bmatrix} G_{11}^v & G_{12}^v & \cdots & G_{1S}^v \\ G_{21}^v & G_{22}^v & \cdots & G_{2S}^v \\ \vdots & \vdots & \vdots & \vdots \\ G_{L1}^v & G_{L2}^v & \cdots & G_{LS}^v \end{bmatrix} \tag{16}$$

where each patch of M_i^t is projected L times with different learned linear projection functions G^q, G^k, and G^v to get queries, keys, and values, respectively. The subscript S denotes that, after linear projection, the features are split into smaller spatial windows in spatial space to facilitate local attention computation, thereby maintaining spatial–channel consistency at the local level. Overall, instead of treating each source equally as query, key, and value like previous works [41], [42], we use P_i^t as the query and the already texture-aligned M_i^t as both key and value, leading to a lightweight and nonredundant structure.

Then, with (11), the inverted attention mechanism is performed as follows:

$$\begin{cases} A_{gqk}^{(l)} = \text{concat}_{\text{space}}\left(\text{concat}_{\text{channel}}\left(A_{gqk_{h,s}}^{(l)}\right)_{h=1}^{H}\right)_{s=1}^{S} \\ V_m{}^{(l)} = \text{concat}_{\text{space}}\left(\text{concat}_{\text{channel}}\left(v_{m_{h,s}}^{(l)}\right)_{h=1}^{H}\right)_{s=1}^{S} \\ F_i = W_f\left(\sum_{l=1}^{L} A_{gqk}^{(l)} \cdot \sum_{l=1}^{L} V_m{}^{(l)}\right) \end{cases} \tag{17}$$

where the subscript of concat. denotes that each group of A_{gqk} is first concatenated along the channel dimension, followed by spatial concatenation. l denotes the number of projected QKV groups. h denotes the number of attention heads per group. s denotes the number of spatial windows. W_f denotes the linear projection.

Self-Attention Exploration: Following the dual-modality feature interaction, the internal dependencies within the fused

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representation are further explored:

$$\begin{cases} A'^{(l)}_{gqk} = \text{concat}_{\text{space}}\left(\text{concat}_{\text{channel}}\left(A'^{(l)}_{gqk_{h,s}}\right)_{h=1}^{H}\right)_{s=1}^{S} \\ V_{F_i}{}^{(l)} = \text{concat}_{\text{space}}\left(\text{concat}_{\text{channel}}\left(v_{F_{i_{h,s}}}{}^{(l)}\right)_{h=1}^{H}\right)_{s=1}^{H} \\ F_i^s = W_f\left(\sum_{l=1}^{L} A'^{(l)}_{gqk} \cdot \sum_{l=1}^{L} v_{F_i}{}^{(l)}\right) \end{cases} \tag{18}$$

where A'_{gqk} is the self-attention map of F_i. Apart from the self-attention exploration between F_i and F_i^s, an spatial–channel attention block is inserted as (6) and (7), to form a secondary branch

$$\begin{cases} F_i^{s'} = F_i^s + F_i^s \odot \sigma\left(\text{Map-S}\left(\varrho\left(F_i\right)\right)\right) \\ F_i' = F_i^{s'} + F_i^{s'} \odot \sigma\left(\text{Map-C}\left(\varrho\left(F_i\right)\right)\right) \end{cases} \tag{19}$$

where $\varrho(\cdot)$ is a convolutional layer. By introducing spatial and channel attention derived from the original fused feature F_i, modality-aware guidance is injected into the internal dependencies of F_i^s. This process enhances both the spatial localization and channel discrimination of F_i^s, resulting in more informative and structured feature representations.

Through the SDFD module, the fusion representations effectively achieve a lightweight form with diversity, modality specificity, and nonredundancy, which lays the foundation for the subsequent classification.

C. DF Classification

The obtained series of fused features F_i' are different in resolution. The smaller the value of i, the more global information F_i' contains. Conversely, it contains richer details and textures. Directly integrating multilevel features, such as concatenation, often leads to conflicts and dilution between the two types of information. To reduce the conflict between different-level features and fuse them up to the final version, we perform a decoupling-based fusion (DF) strategy

$$F_{out} = \mathcal{D}\left(\cdots \mathcal{D}\left(\mathcal{D}\left(F_1', F_2'\right), F_3'\right) \cdots, F_n'\right). \tag{20}$$

The function $z = \mathcal{D}(x, y)$ serves for feature decoupling and integration, which is composed of two parts of calculations.

Feature Decoupling: First, for x, which contains more global modeling information than y

$$\begin{cases} X = \text{FlowMake}\left(x\right) \\ B\left(p_s\right) = \sum_{p \in \mathcal{N}} w_p X\left(p\right) \\ E = F_i' - B \end{cases} \tag{21}$$

the flow field generation module is used to generate the flow field, which is described as X. Then, X is used to generate the foreground body component B, and E furthermore.

Here, X represents the flow field, B represents the foreground body component, and E represents the details and boundary information. p_s refers to the points in B. p is any point in the feature space X. $\mathcal{N}$ represents the set of points surrounding p. w_p denotes the bilinear kernel weights on the warped spatial grid.

Feature Integration: After decoupling the boundary information E, it is concatenated with y, which contains more boundaries and textures. This combined data is then enhanced using a convolution block. Finally, the boundary and texture features are recombined with the foreground component to obtain the final fused feature

$$z = B + \mathcal{C}\left(\text{concat}\left(E, y\right)\right) \tag{22}$$

where $\mathcal{C}(\cdot)$ denotes a convolutional layer. In the end, F_{out} is input into the fully connected classification layer. The loss function is the cross-entropy function

$$\text{loss} = -\frac{1}{N}\sum_{i=1}^{N}\left[F_{out_i}\log\left(\hat{F}_{out_i}\right) + \left(1 - F_{out_i}\right)\log\left(1 - \hat{F}_{out_i}\right)\right]. \tag{23}$$

IV. Experimental Results

In the following, we introduce the dataset, experimental setup, and the experimental results that ultimately demonstrate the effectiveness of our proposed method.

A. Datasets Discription

We evaluate the proposed method on three datasets comprising PAN and MS images from the GF-2 satellite. Ground truth (GT) is annotated from the MS data.

1) Hohhot Dataset: As illustrated in Fig. 4(a), the Hohhot dataset includes 11 manually labeled categories (GT1–GT11), specifically: Water, Road, Bare Land, Vegetation, Intensive Building, Sparse Building, Suburb Building, Farmland, White Building, Red Building, and Blue Building. The MS data have four spectral bands: red, green, blue, and near-infrared. Its spatial resolution is $2001 \times 2001 \times 4$. The PAN data comprise a single channel with a resolution of $8004 \times 8004 \times 1$.

2) Nanjing Dataset: As presented in Fig. 4(b), the Nanjing dataset contains 11 categories (GT1–GT11), which are: Water, Road1, Road2, Bare Land, Vegetation1, Vegetation2, Building1red, Building2blue, Building3intensive, Building4white, and Building5sparse. The MS data dimensions are $2000 \times 2500 \times 4$ pixels. The PAN data dimensions are $8000 \times 10\,000 \times 1$ pixels.

3) Xi'an Dataset: As illustrated in Fig. 4(c), Xi'an dataset contains 12 categories (GT1–GT12), namely, Water, Road1, Road2, Bare Land, Low Vegetation, Farmland, Tree, Intensive Building, Sparse Building, Suburb Building, Workshop Building, and White Building. The MS data have a resolution of $4548 \times 4544 \times 4$ pixels. The PAN data are sized at $18\,192 \times 18\,164 \times 1$ pixels.

B. Experimental Settings and Environment

We partition the dataset into different patches to serve as inputs for the network. For each pixel in the MS data, we extract a patch of specific size centered on this pixel. Corresponding patches of four times the size of MS patches are extracted from the PAN data to maintain spatial alignment. Under the linear scale-space theory, we choose to use an exponential scale with a base of 2 to implement a continuously varying scale space. The series of input image groups $group_i$

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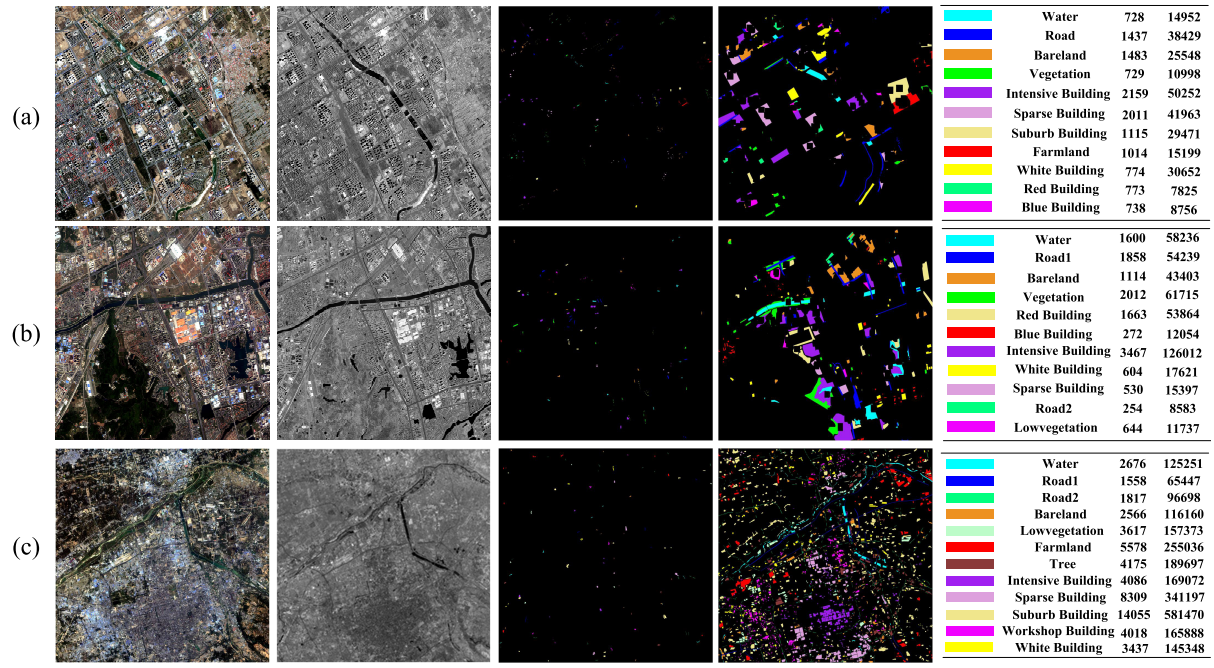

Fig. 4. Three rows are the datasets used. (a) Hohhot dataset, (b) Nanjing dataset, and (c) Xi'an dataset. The first column is the MS set, the second column is the PAN set, the third column is the train set, the fourth column is the GT of the test set, and the last column is the number of each class in the datasets.

are set with resolutions of 1x × 1x, 2x × 2x, and 4x × 4x. The initial embedding dimension of each group in the tokenizer is set to 48. In SDFD, the number of heads in the multihead attention mechanism is configured to 6, and the split-window size is set to x/2× x. The number of groups in the internal dependencies exploration (18) is set to 3. r_1 and r_2 in (6) and (7) are set to 2 and 8, respectively.

To prevent data leakage, no pixel overlap exists between the training and testing sets. The training process is conducted with a fixed learning rate of 0.0001 and a batch size of 42 for 40 epochs when all the models have reached convergence. The methods to evaluate the performance of our proposed method are three metrics: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KAPPA). In addition, the time metrics T_1 refer to the training time. T_2 refers to the testing time.

To guarantee the reliability of our proposed method, we perform five independent experiments on the datasets and calculate the average results for the final performance assessment. All experiments are executed on a system with an RTX 3090 GPU (24 GB) and 128 GB of RAM, running on Ubuntu 22.04 and the PyTorch framework.

C. Comparison and Analysis of Hyperparameters

In this section, we employ the method of controlling variables to compare and analyze the impact of different hyperparameter values on the classification results.

1) Number of Groups in Dual-Source Feature Interaction L: As shown in (17), L represents the number of groups into which the dual-source feature is mapped as queries, keys, and values. We set the patch sizes of a series of input features to 16×16, 32×32, and 64×64, and conducted experiments with L set to 1–3. The experimental results on the three datasets are illustrated in Table I. Clearly, when $L = 2$, the overall OA of all three datasets reaches its highest level. When $L = 1$ or 3, the OA of all three datasets is lower than that when $L = 2$, and the OA when $L = 3$ is lower than that when $L = 1$. We speculate that the model overfits when $L > 2$.

TABLE I
EXPERIMENTAL RESULTS FOR DIFFERENT L FACTORS

l	dataset	1	2	3
OA(%)	Hohhot	73.28	**75.19**	72.83
	Nanjing	77.86	**79.02**	71.96
	Xi'an	80.19	**80.64**	78.20

TABLE II
EXPERIMENTAL RESULTS FOR DIFFERENT PATCH SIZES

PATCH SIZE	dataset	8/16/32	16/32/64	32/64/128
OA(%)	Hohhot	68.02	75.19	**76.76**
	Nanjing	74.09	**79.02**	73.73
	Xi'an	77.17	**80.64**	72.65

2) Patch Size of the Series of Input Image Groups 1x/2x/4x: As mentioned in the experiment settings, x stands for the minimum resolution cardinality. Based on the results gained from the above experiments, we set the number of groups L to 2. Given the scale range of land covers and the constraints of GPU memory, we set x to 8, 16, and 32. The experimental results on the three datasets are illustrated in Table II. When the patch sizes of the input feature series are 16/32/64, the OA of Nanjing and Xi'an datasets reaches its highest level. When the patch size increases or decreases, the OA decreases. For the Hohhot dataset, as the patch sizes of the input feature series increase from 8/16/32 to 16/32/64, the OA significantly improves. When the patch sizes further increase from 16/32/64

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TABLE III
EXPERIMENT RESULTS OF EACH COMPONENT ON THE HOHHOT DATASET

Baseline	TMFG	SDFD	DF	OA(%)	AA(%)	KAPPA(%)
✓				72.86	76.62	69.47
✓	✓			74.22	79.50	71.13
✓	✓	✓		74.14	79.41	71.02
✓	✓		✓	69.58	77.88	66.21
✓	✓	✓	✓	**76.76**	**81.33**	**73.90**

TABLE IV
EXPERIMENTAL RESULTS OF BASELINE AND TMFG ON THE HOHHOT DATASET

Category	Baseline	Baseline + TMFG
1: Water	**98.63**	95.72
2: Road	**80.78**	80.02
3: Bareland	78.77	**84.68**
4: Vegetation	**65.70**	53.66
5: Intensive Building	79.32	**89.25**
6: Sparse Building	61.14	**71.66**
7: Suburb Building	97.26	**99.06**
8: Farmland	84.18	**88.92**
9: White Building	**63.33**	57.80
10: Red Building	45.05	**47.55**
11: Blue Building	**37.66**	37.25
OA	72.86	**74.22**
AA	76.62	**79.50**
Kappa	69.47	**71.13**

to 32/64/128, the OA slightly improves, ultimately reaching its highest level.

D. Ablation Study

In this section, we take the Hohhot dataset as an example to analyze the overall contributions of different modules. ResNet-18 is used as the baseline. Also, simple weighted sum and concatenation are used for its fusion classification. The results are shown in Table III. Compared to the baseline, baseline + TMFG shows an improvement of 1.36% in OA, indicating high effectiveness. However, baseline + TMFG + SDFD module results in a slight decrease of 0.07% in OA. Compared with baseline + TMFG, baseline + TMFG + DF causes OA to drop below the baseline by 4.64%. This is because the information obtained by SDFD is multilevel. Simple weighted fusion may lead to dilution and redundancy. DF is also more effective in integrating compact and informative features extracted by SDFD, but shows limited benefit for redundant features. Finally, baseline + TMFG + SDFD + DF contributes the most to the overall network, increasing the OA by 3.90% compared to the baseline. This demonstrates the collaborative contribution of all modules, especially the effective cooperation between SDFD and DF.

E. Module Comparison Experiments

In this section, we use the Hohhot dataset as an example to further verify the function of TMFG and SDFD. Three pairs of comparative experiments are conducted for validation.

TABLE V
QUANTITATIVE RESULTS ON HOHHOT DATASET

Models	NNCNet	MFT	AM³Net	CRHFF	GCFnet	ISSPNet	SNCFNet	OURS
c_1	96.32	**97.27**	96.25	96.99	85.46	92.45	95.21	96.05
c_2	78.89	79.84	**94.27**	82.58	67.35	82.82	84.61	83.43
c_3	80.45	82.28	86.13	87.25	65.97	84.01	**87.92**	84.52
c_4	59.50	71.08	58.74	**87.95**	49.70	79.24	80.81	66.93
c_5	74.27	68.06	80.44	63.17	65.86	63.79	59.59	**88.66**
c_6	60.18	59.05	55.30	59.96	51.01	64.18	65.43	**70.21**
c_7	93.52	86.09	96.16	65.49	86.87	71.92	77.31	**99.67**
c_8	82.69	76.36	49.61	94.34	72.54	**95.12**	93.37	81.27
c_9	66.01	64.57	65.48	59.61	66.26	62.65	**66.37**	61.34
c_{10}	52.21	48.72	54.03	**75.50**	38.20	72.32	73.16	64.80
c_{11}	57.70	41.22	53.92	75.39	31.31	**78.13**	73.61	43.01
OA(%)	73.42	70.79	72.65	72.64	63.81	73.77	74.80	**76.76**
AA(%)	76.30	73.43	71.85	77.11	63.35	76.97	77.94	**81.33**
Kappa(%)	69.98	66.99	69.55	69.46	59.06	70.39	71.56	**73.90**
T_1(s)	712.26	**200.44**	442.72	1280.10	747.96	883.25	2204.49	1998.40
T_2(s)	102.21	**40.74**	81.10	212.65	136.14	157.11	687.50	135.23
Params(M)	2.22	**0.21**	2.78	4.91	3.90	12.08	6.50	2.38

TABLE VI
QUANTITATIVE RESULTS ON NANJING DATASET

Models	NNCNet	MFT	AM³Net	CRHFF	GCFnet	ISSPNet	SNCFNet	OURS
c_1	93.63	84.12	89.81	**96.44**	88.72	86.52	90.72	86.77
c_2	65.74	61.12	45.82	**73.22**	47.83	40.13	63.29	65.80
c_3	87.39	73.13	79.78	**95.43**	53.81	53.73	65.79	72.86
c_4	76.97	77.59	65.44	**84.28**	77.11	72.18	81.66	83.24
c_5	77.94	88.80	88.26	77.18	37.90	61.30	88.20	**93.28**
c_6	90.20	89.67	93.14	93.27	40.17	81.52	77.47	**96.07**
c_7	75.46	86.21	80.26	58.69	63.01	75.64	85.73	**87.56**
c_8	84.80	84.65	61.89	84.08	79.79	**88.43**	80.55	87.07
c_9	48.05	58.96	39.67	50.73	32.58	43.67	**61.40**	55.87
c_{10}	58.86	28.14	**87.02**	46.06	46.17	72.67	43.37	28.47
c_{11}	**67.82**	65.32	57.22	65.91	42.60	22.64	55.83	55.42
OA(%)	76.34	77.35	72.22	75.39	61.26	63.50	78.26	**79.02**
AA(%)	75.16	71.66	74.38	75.03	52.34	59.56	72.87	**78.54**
Kappa(%)	73.53	73.70	68.37	71.54	54.53	57.72	74.62	**75.65**
T_1(s)	720.54	**246.50**	456.84	1366.92	521.92	1000.89	2279.87	1800.18
T_2(s)	173.79	**57.43**	141.45	246.87	235.77	281.71	1149.63	227.31
Params(M)	2.22	**0.21**	2.78	4.91	3.90	12.08	6.50	2.38

1) Analysis of TMFG Module: As shown in Table IV, we first compared the results of the baseline model and baseline + TMFG. The integration of TMFG leads to a notable improvement in OA, AA, and KAPPA. Class-wise analysis reveals that TMFG improves the accuracy of some classes by more than 4%, such as c_3, c_5, c_6, and c_8 that exhibit complex textures and varying scales. In addition, the baseline + TMFG model outperforms all SOTA methods mentioned in Table V in terms of OA, AA, and KAPPA, further validating the effectiveness and generalizability.

Moreover, a comparative experiment is conducted between our best-performing model and a modified model that is constructed by substituting the TMFG of the best-performing model with conventional multiscale methods. The visualizations of their intermediate results are presented in Fig. 5(b1) and (b2). Compared to the original image in (a), it is evident that Fig. 5(b2) features are clearer than (b1) with enhanced

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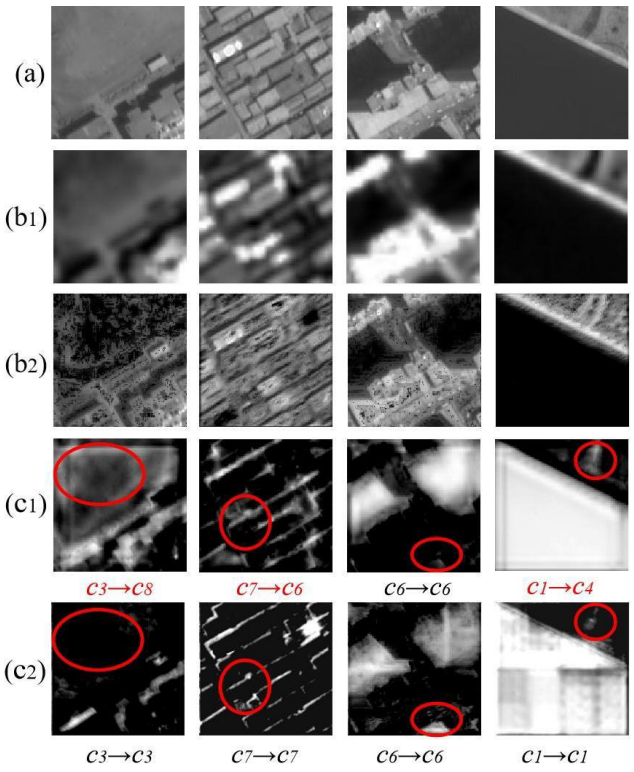

Fig. 5. Three sets of feature maps are demonstrated here, and each set is composed of four distinct regions. Images of the same region are placed in one column. (a) Origin feature maps. (b1) and (b2) Pair of intermediate features from comparative experiments mentioned in the second paragraph of Section IV-E1. (b1) Best-performing model. (b2) Modified model. (c1) and (c2) Pair of intermediate features from comparative experiments mentioned in Section IV-E2. (c1) Best-performing model. (c2) Separation-fusion pattern model. The numbers below them show the detailed classification results.

texture details; (b1) features appear blurred, and certain patterns exhibit spatial misalignment. These observations indicate that our model achieves superior performance in constructing texture-aligned multiscale representations.

2) Analysis of SDFD Module: In this pair of experiments, we compared our best-performing model with a separation-fusion pattern model based on multiscale Transformer [41]. We visualize the differences between their respective final feature representations in Fig. 5(c1) and (c2). As observed, features in Fig. 5(c1) are more blurred compared to (c2). Fig. 5(c2) features capture salient patterns of various categories more accurately and reinforce modality-specific characteristics more effectively. For example, in the second column, Suburb Building class, our model highlights the texture of suburb buildings well, whereas Fig. 5(c1) produces blurry features with partial texture loss, leading to misclassification. In the fourth column, Water class, both models attempt to emphasize water regions, but Fig. 5(c1) incorrectly highlights a nonwater region within inland areas, resulting in misclassification. However, Fig. 5(c1) contains a black linear noise pattern in the lower part, which may be caused by the split-window strategy of SDFD.

Overall, SDFD effectively generates more compact and modality-specific features, thereby addressing the sparsity and redundancy problem introduced by separation-fusion strategies.

F. Comparison With Other Methods

In this section, we compare our proposed MSAFormer with other state-of-the-art (SOTA) methods on three datasets to validate its effectiveness. These SOTA methods have been illustrated in Section I, which are NNCNet, MFT, AM^{3}Net, CRHFF, GCFNet, ISSP-Net, and SNCF-Net. Their parameter settings correspond to their original papers to ensure optimal performance. The evaluations are based on OA, AA, and KAPPA.

1) Experimental Result on Hohhot Dataset: The quantitative experiment results of different methods on the Hohhot dataset are shown in Table V and Fig. 6. The following content is the analysis of our experimental results.

NNCNet proposes nearest neighbor augmentation and bilinear attention and noticeably better classifies c_9 (White Building) and c_{11} (Blue Building), benefiting from the exploitation of local semantic relationships. In contrast, our Transformer-based SDFD focuses on long-range dependencies, occasionally misclassifying local pixels. Overall our results surpasses NNCNet on c_2–c_7, c_{10}, and all three metrics, demonstrating outstanding performance.

MFT mainly exploits spectral features with LiDAR as auxiliary input, yielding strong performance on c_1 (Water) and performs relatively evenly across other categories. In contrast, our method performs better on c_5 (Intensive Building), c_6 (Sparse Building), and c_7 (Suburb Building) by effectively modeling spatial information through the texture-aware multiresolution space and compact feature interaction. Overall, our approach surpasses MFT across all three metrics, demonstrating more balanced performance.

AM^{3}Net integrates spectral and spatial cues through mutual-guided modules and involution operators. Compared to our method, it performs significantly better on c_2, c_9, and c_{11} for its focus on pixel-level spectral features. Our method, however, excels on c_5–c_8, where long-range spectral and spatial cues matter, showing a stronger balance between the two. Overall, we achieve higher OA, AA, and KAPPA than AM^{3}Net, indicating the effectiveness of our method.

CRHFF establishes an autoencoder-like network from MS to PAN, enabling the fusion of features at deep cross-resolution hidden layers. Compared to our method, it exhibits great advantages on c_4, c_8, c_{10}, and c_{11}. Our method outperforms much higher on c_5–c_7 and c_9. Finally, our OA, AA, and KAPPA are higher than those of CRHFF. The reason is quite similar to AM^{3}Net. Note that our accuracy on c_{10} and c_{11} is significantly lower than others, which may be due to the insufficient attention to local texture cues and oversensitivity to local spectral features. The limited number of training samples may also be one of the contributing factors.

GCFNet is a patch-free method based on an encoder–decoder network. Compared to our method, except for c_9, the performances of other classes and the three evaluation metrics are all lower than our method. On the one hand, the datasets we use are substantially more extensive, encompassing a greater diversity of complex terrestrial features, thereby enriching the depth of our model. On the other hand, our method incorporates a more intricate design in hierarchical multiresolution construction

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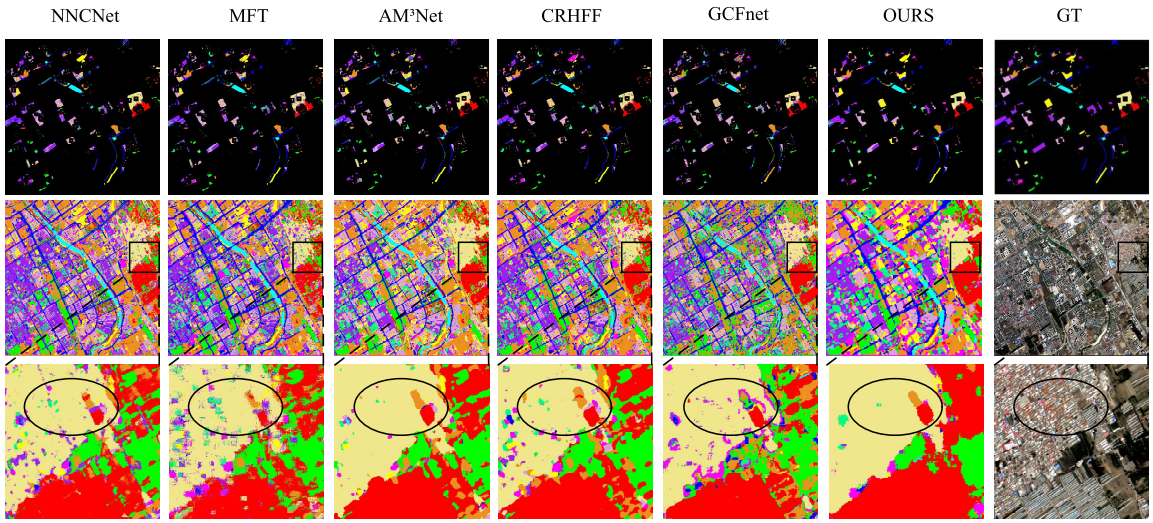

Fig. 6. Classification maps for the test set and the complete image of the Hohhot dataset. The first row displays the classification outcomes for the test set. The second row illustrates the classification outcomes for the complete image. The final row provides a detailed view of a specific section from the second row's image. The last column represents the GT map alongside the MS visualization.

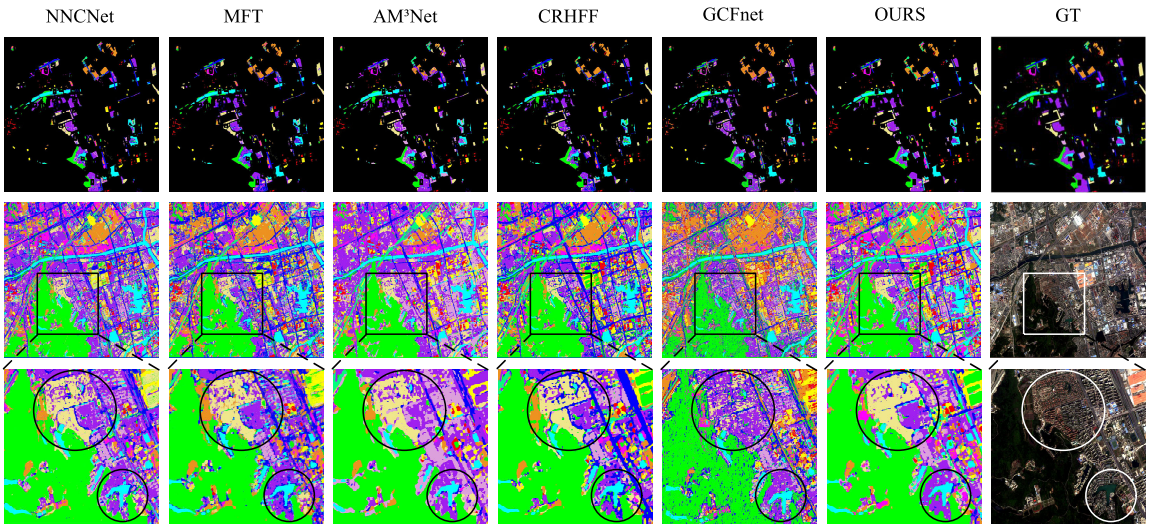

Fig. 7. Classification maps for the test set and the complete image of Nanjing dataset. The first row displays the classification outcomes for the test set. The second row illustrates the classification outcomes for the complete image. The final row provides a detailed view of a specific section from the second row's image. The last column represents the GT map alongside the MS visualization.

and generates a more compact and comprehensive fusion for classification.

ISSP-Net introduces a pixel-guided multiscale spatial module and a time–frequency spectral module to extract complementary information. It performs considerably better on c_8, c_{10}, and c_{11}, which owes to its pixel-level design. However, its complex design also results in a larger number of training parameters. Nevertheless, our method outperforms ISSP-Net in c_1–c_3, especially in c_5–c_7. This is mainly due to our texture-aware multiresolution Transformer and the nonredundant feature fusion measure. Overall, our method achieves superior performance across all three evaluation metrics, demonstrating its balanced effectiveness.

SNCF-Net proposes a network based on continuous-scale inputs and a response convolution kernel to achieve feature interaction. It noticeably outperforms our method on c_3, c_4, and c_8–c_{11}, which reflects the superiority of their continuous-scale inputs and convolutional architecture in capturing discriminative local information. However, its complex design results in a larger number of training parameters and longer training and testing time. In contrast, our Transformer-based design achieves superior results on c_1 and c_5–c_7 by modeling global dependencies well, and surpasses SNCF-Net across all three metrics.

2) Experimental Result on Nanjing Dataset: The quantitative experimental results of different methods on the Nanjing dataset are shown in Table VI and Fig. 7. We achieve the highest classification accuracy on c_5 (Building1red), c_6 (Building2blue), and c_7 (Intensive Building). CRHFF achieves the highest accuracy on four classes. However, our method achieves the best performance in OA, AA, and KAPPA, indicating that we have more balanced and consistent performance across all classes. The OA, AA, and KAPPA of SNCF-Net are also very competitive.

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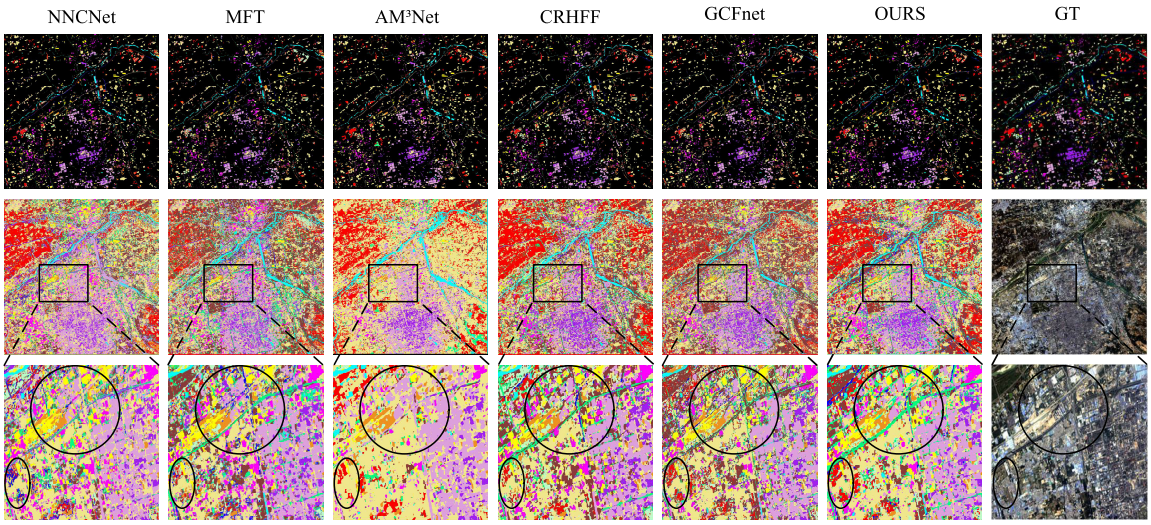

Fig. 8. Classification maps for the test set and the complete image of Xi'an dataset. The first row displays the classification outcomes for the test set. The second row illustrates the classification outcomes for the complete image. The final row provides a detailed view of a specific section from the second row's image. The last column represents the GT map alongside the MS visualization.

TABLE VII
QUANTITIVE RESULTS ON XI'AN DATASET

Models	NNCNet	MFT	AM³Net	CRHFF	GCFnet	ISSPNet	SNCFNet	OURS
c_1	93.89	92.33	95.45	93.05	90.81	96.75	**96.90**	93.76
c_2	33.15	42.78	71.71	25.55	39.70	26.29	26.84	**50.80**
c_3	61.04	50.96	60.13	62.11	63.52	**68.75**	62.17	53.43
c_4	69.70	68.33	65.44	67.34	72.42	72.70	69.83	**77.68**
c_5	**58.32**	44.42	52.33	54.23	32.59	54.81	45.09	48.47
c_6	69.46	75.28	80.41	72.09	77.34	71.42	75.11	**86.22**
c_7	77.49	70.85	**86.62**	84.46	67.11	86.47	86.27	64.46
c_8	74.09	**86.47**	61.46	77.97	64.48	74.57	77.09	75.62
c_9	78.89	83.88	61.96	79.47	**89.06**	77.17	76.63	83.75
c_{10}	94.40	95.91	82.92	97.13	84.11	95.47	**97.88**	91.75
c_{11}	89.19	84.17	95.72	**96.38**	88.15	92.03	94.56	94.66
c_{12}	78.06	72.87	**96.35**	70.38	88.77	67.29	76.38	88.92
OA(%)	78.32	76.77	75.38	79.69	76.20	79.58	80.34	**80.64**
AA(%)	73.14	68.75	65.35	73.34	68.35	73.64	73.73	**74.64**
Kappa(%)	75.29	73.41	71.60	76.82	72.77	76.66	77.51	**77.94**
T_1(s)	2880.22	**845.92**	1648.46	4386.66	2934.1	3800.29	8160.96	7196.28
T_2(s)	1099.35	**580.64**	943.72	1692.48	1433.39	1946.86	6017.70	1437.28
Params(M)	2.22	**0.21**	2.78	4.91	3.90	12.08	6.50	2.38

3) Experimental Result on Xi'an Dataset: The quantitative experimental results of different methods on the Xi'an dataset are shown in Table VII and Fig. 8. We perform the best on c_2 (Road1), c_4 (Bareland), and c_6 (Farmland). AM^{3}Net and SNCF-Net achieve the highest accuracy on two classes. SNCF-Net achieves the highest accuracy on c_1 and c_{10}, but the gaps between our method and SNCF-Net are small. The OA, AA, and KAPPA of SNCF-Net are also very competitive. Overall, our method achieves the highest results in OA, AA, and KAPPA.

V. CONCLUSION

In this article, we introduce an MSAFormer for PAN and MS joint classification, achieving competitive results on three datasets. Experimental results demonstrate that TMFG exhibits strong capability in constructing precise texture-aligned multiscale foundations, effectively protecting the detailed information of original inputs. This semantic alignment approach provides a fresh perspective for multiscale feature alignment in multimodal remote sensing imagery. The SDFD module generates compact and comprehensive fusion representations, which capture enhanced modality-specific features and eliminate the overlapped redundancy of shared information in a nondecoupled manner. However, we observe that the SDFD module may be insensitive to local texture cues during interaction and oversensitive to local spectral information, which affects the accuracy of small spectrally distinctive categories. Also, the split-window design in SDFD may sometimes cause linear noise. In the future, we plan to explore a more elaborate local attention to better balance spectral–spatial feature modeling.

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